

Where does space begin?

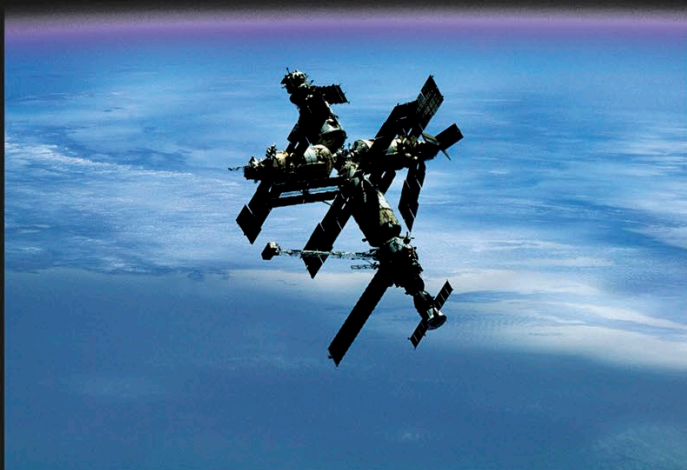
The Earth is cloaked in a thin layer of gases – the atmosphere. Outside this atmosphere is space, where there is no air to breathe, or to allow wings to fly, and where nobody can hear you scream.

Fading away

Our atmosphere does not just end suddenly – it fades gradually into space.

View from Mir

The Russian Mir space station was in orbit for 15 years. Here, it was photographed by the US shuttle *Atlantis*.



EXOSPHERE



The exosphere is the outer layer of the atmosphere, extending about 10,000 km (6,000 miles) above the ground. From here, lighter gases drift into space beyond.

THERMOSPHERE



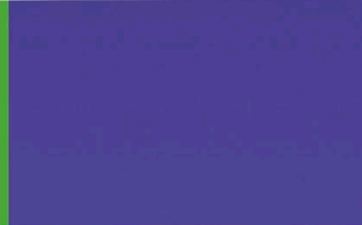
The thermosphere reaches way up to more than 700 km (over 400 miles) above the Earth. The polar lights (aurora borealis in the north and aurora australis in the south) glow in the thermosphere.

MESOSPHERE



Most experts agree that space begins at 100 km (63 miles) above the ground. Past this, our image is not drawn to scale.

STRATOSPHERE



The mesosphere extends about 85 km (53 miles) above the ground. The air is thin here, but it's still thick enough to slow meteorites down.

TROPOSPHERE



The stratosphere rises about 50 km (31 miles) above the Earth. Planes cruise in the upper troposphere or lower stratosphere, above the clouds.

The troposphere extends between 6 and 20 km (3½–12 miles) above the ground. All our weather takes place in the troposphere.